

From separate sensors to PID and fusion evaluation

Proposed offline workflow · fixed context · one declared physical target

1 Raw observations

Camera 1
RGB: $H \times W \times 3$
or grayscale: $H \times W \times 1$

Camera 2
Its own frame + time
View + calibration identity

Pressure microphone
Window ending by t
Rate · units · clipping · age

2 Separate extractors

Frozen vision extractor
This camera; context fixed
Small feature vector z_1

Frozen vision extractor
This camera; context fixed
Small feature vector z_2

Declared audio extractor
RMS, bands, or frozen model
Small feature vector z_3

3 Categorical sources

Declared bins $\rightarrow X_1$
One group per camera
Retain the encoding identity

Declared bins $\rightarrow X_2$
Keep the two views separate
Shared weights $\neq$ fused inputs

Declared bins $\rightarrow X_3$
One group per audio view
Loudness is not direction

Split whole episodes before fitting or tuning. Freeze all selected maps.

Keep sensor ID, timestamps, fixed context, availability M and quality Q .

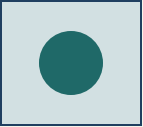
4 Aligned features z , categories X , and reference Y
Reference Y comes from a separate measurement / annotation route.

5a Offline information analysis
Existing Rust: categorical MGW, 2–4 sources
Signed atoms + subset MI / CMI
Describes the retained feature categories

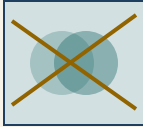
5b Separate fusion-model study
Train on X or retained features z
Compare all 8 subsets for 3 sources
Held-out loss · outages · calibration · cost

Quantized-feature PID does not characterize all raw pixels, audio, or a richer model input.

Quality is a separate record, not a binary availability flag

**Usable**
 $M = 1$
Reference condition

**Partial**
 $M = 1$
Object partly hidden

**Distorted**
 $M = 1$
Blur, noise, or clipping

**Missing**
 $M = 0$
No observation

Compare supported quality regimes on common episodes and fixed encodings.

An offline quality label is not automatically available to either predictor.

Natural occlusion can depend on the target. Do not assume random dropout.